

SET A

```
public class Anime {

    String title;
    String genre = "Unknown";
    Anime previous;

    String[] episodeNames = new String[20];
    int[] durations = new int[20];
    boolean[] canon = new boolean[20];
    double[] ratings = new double[20];
    int count = 0;

    public Anime(String title) {
        this.title = title;
    }

    public Anime(String title, String genre) {
        this.title = title;
        this.genre = genre;
    }

    public void addEpisode(String name) {
        addEpisode(name, 18, true);
    }

    public void addEpisode(String name, int duration, boolean isCanon) {
        episodeNames[count] = name;
        durations[count] = duration;
        canon[count] = isCanon;
        ratings[count] = 0.0;

        System.out.println("Added: " + name + " | Duration: " +
            duration + " min | Canon: " +
            (isCanon ? "Yes" : "No"));

        count++;
    }

    public void setPrevious(Anime a) {
        previous = a;
    }

    public void rateEpisode(String name, double rating) {
        for (int i = 0; i < count; i++) {
            if (episodeNames[i].equals(name)) {
                ratings[i] = rating;
                break;
            }
        }
    }
}
```

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    }
}

public void display() {
    System.out.print("Anime: " + title +
        " | Genre: " + genre +
        " | Previous: ");

    if (previous == null)
        System.out.println("None");
    else
        System.out.println(previous.title);
    for (int i = 0; i < count; i++) {
        System.out.println(" [" + (i + 1) + "] " +
            episodeNames[i] +
            " | Rating: " + ratings[i] + "/10");
    }
}

public void canonDuration() {
    int total = 0;

    for (int i = 0; i < count; i++) {
        if (canon[i]) {
            total += durations[i];
        }
    }
    System.out.println("Total Canon Duration of " + title +
        ": " + total + " min");
}

public void topRated() {
    int max = 0;
    for (int i = 1; i < count; i++) {
        if (ratings[i] > ratings[max]) {
            max = i;
        }
    }
    System.out.println("Top Rated Episode of " + title + ": " +
        episodeNames[max] + " (" + ratings[max] + "/10)");
}
}

```

SET B

```
public class Anime {

    String title;
    String genre = "Unknown";
    Anime previous;

    String[] episodeNames = new String[20];
    int[] durations = new int[20];
    boolean[] canon = new boolean[20];
    double[] ratings = new double[20];
    int count = 0;

    public Anime(String title) {
        this.title = title;
    }

    public Anime(String title, String genre) {
        this.title = title;
        this.genre = genre;
    }

    public void addEpisode(String name) {
        addEpisode(name, 20, false);
    }

    public void addEpisode(String name, int duration, boolean isCanon) {
        episodeNames[count] = name;
        durations[count] = duration;
        canon[count] = isCanon;
        ratings[count] = 0.0;
        System.out.println("Added: " + name +
            " | Duration: " + duration +
            " min | Canon: " + (isCanon ? "Yes" : "No"));
        count++;
    }

    public void setPrevious(Anime a) {
        previous = a;
    }

    public void rateEpisode(String name, double rating) {
        for (int i = 0; i < count; i++) {
            if (episodeNames[i].equals(name)) {
                ratings[i] = rating;
                return;
            }
        }
    }
}
```

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public void display() {
    System.out.print("Anime: " + title +
        " | Genre: " + genre +
        " | Previous: ");

    if (previous == null)
        System.out.println("None");
    else
        System.out.println(previous.title);
    for (int i = 0; i < count; i++) {
        System.out.println(" [" + (i + 1) + "] " +
            episodeNames[i] +
            " | Rating: " + ratings[i] + "/10");
    }
}

public void canonDuration() {
    int total = 0;
    for (int i = 0; i < count; i++) {
        if (canon[i]) {
            total += durations[i];
        }
    }
    System.out.println("Total Canon Duration of " + title + ": " + total + " min");
}

public void topRated() {
    int max = 0;
    for (int i = 1; i < count; i++) {
        if (ratings[i] > ratings[max]) {
            max = i;
        }
    }
    System.out.println("Top Rated Episode of " + title + ": " +
        episodeNames[max] + " (" + ratings[max] + "/10)");
}
}

```

RUBRICS:

Correct instance variables	- 1 mark
Constructor Overloading	- 1 mark
addEpisode() method overloading	- 0.5 mark for separate method

	- 1.5 marks for updating variables & printing
display()	- 1.5 marks correct printing
setPrevious()	- Object reference 1 mark
rateEpisode()	- Update correct anime by name 1 mark
canonDuration()	- Calculate total canon duration iteratively 1.5 marks
topRated()	- Correct maximum rated episode 1 mark